DESCENT OF A BIVARIANT CLASS ALONG AN ENVELOPE FROM THE EQUALITY OF ITS TWO RESTRICTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source line 6821, 'The other conditions are proved in the same manner and we omit the detailed arguments.', is retained verbatim at the end of the counted proof. Condition (1) of the bivariant class definition is carried out in full inside the span (lines 6784-6820); only the two remaining conditions of the same routine argument are left to the reader, which is the routine-verification residue that LANE_BRIEF section 5 accepts.

Allowed standard prerequisites (editorial).

- Right exactness of the Chow sequence of an envelope. Same chapter chow.tex, label lemma-envelope, invoked at proof line 6737. Supplies that every beta in CH_k(Y) lifts to beta' in CH_k(Y') with g_* beta' = beta, which is what makes the defining formula c cap beta = h_*(c' cap beta') possible; the rest of the argument is in the counted span.
- Restriction of a bivariant class along a cartesian square. Same chapter chow.tex, label remark-restriction-bivariant, invoked at proof lines 6732-6735. Identifies res_1(c') and res_2(c') with the restrictions for the two cartesian squares of the displayed diagram, which is the hypothesis consumed at proof line 6736.
- The axioms of a bivariant class and base change of envelopes. Same chapter chow.tex, labels definition-bivariant-class (proof lines 6762 and 6782) and lemma-base-change-envelope (proof line 6766); definition-envelope (proof line 6710). Fix the three conditions that the descended operation must satisfy and guarantee that the base change Y'_1 -> Y_1 of an envelope is again an envelope, so the same lift can be performed after any locally finite type base change.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0B76: sources/stacks-chow/pinned-original-source.tex, source lines 6087–6087. Navigation pointer for \ref{definition-bivariant-class} used inside the counted span (line 6763); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GU5: sources/stacks-chow/pinned-original-source.tex, source lines 3553–3553. Navigation pointer for \ref{definition-envelope} used inside the counted span (line 6710); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GU7: sources/stacks-chow/pinned-original-source.tex, source lines 3580–3580. Navigation pointer for \ref{lemma-base-change-envelope} used inside the counted span (line 6766); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GU8: sources/stacks-chow/pinned-original-source.tex, source lines 3593–3593. Navigation pointer for \ref{lemma-envelope} used inside the counted span (line 6737); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Navigation pointer for \ref{remark-restriction-bivariant} used inside the counted span (line 6733); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 6708); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a morphism of schemes locally of finite type over S. Let $g : Y' \to Y$ be an envelope (Definition 0GU5) and denote $X' = Y' \times_Y X$. Let $p \in \mathbf{Z}$ and let $c' \in A^p(X' \to Y')$. If the two restrictions*

$$res_1(c') = res_2(c') \in A^p(X' \times_X X' \to Y' \times_Y Y')$$

are equal (see proof), then there exists a unique $c \in A^p(X \to Y)$ whose restriction $res(c) = c'$ in $A^p(X' \to Y')$.

Proof. We have a commutative diagram

$$\begin{array}{ccccc} X' \times_X X' & \xrightarrow[b]{a} & X' & \xrightarrow{h} & X \\ \downarrow{\scriptstyle f''} & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y' \times_Y Y' & \xrightarrow[q]{p} & Y' & \xrightarrow{g} & Y \end{array}$$

The element $res_1(c')$ is the restriction (see Remark 0F9Z) of c' for the cartesian square with morphisms a, f', p, f'' and the element $res_2(c')$ is the restriction of c' for the cartesian square with morphisms b, f', q, f''. Assume $res_1(c') = res_2(c')$ and let $\beta \in \mathrm{CH}_k(Y)$. By Lemma 0GU8 we can find a $\beta' \in \mathrm{CH}_k(Y')$ with $g_*\beta' = \beta$. Then we set

$$c \cap \beta = h_*(c' \cap \beta')$$

To see that this is independent of the choice of β' it suffices to show that $h_*(c' \cap (p_*\gamma - q_*\gamma))$ is zero for $\gamma \in \mathrm{CH}_k(Y' \times_Y Y')$. Since c' is a bivariant class we have

$$h_*(c' \cap (p_*\gamma - q_*\gamma)) = h_*(a_*(c' \cap \gamma) - b_*(c' \cap \gamma)) = 0$$

the last equality since $h_* \circ a_* = h_* \circ b_*$ as $h \circ a = h \circ b$.

Observe that our choice for $c \cap \beta$ is forced by the requirement that $res(c) = c'$ and the compatibility of bivariant classes with proper pushforward.

Of course, in order to define the bivariant class c we need to construct maps $c \cap - : \mathrm{CH}_k(Y_1) \to \mathrm{CH}_{k+p}(Y_1 \times_Y X)$ for any morphism $Y_1 \to Y$ locally of finite type satisfying the conditions listed in Definition 0B76. Denote $Y_1' = Y' \times_Y Y_1$, $X_1 = X \times_Y Y_1$. The morphism $Y_1' \to Y_1$ is an envelope by Lemma 0GU7. Hence we can use the base changed diagram

$$\begin{array}{ccccc} X_1' \times_{X_1} X_1' & \overset{a_1}{\underset{b_1}{\rightrightarrows}} & X_1' & \xrightarrow{h_1} & X_1 \\ \downarrow{\scriptstyle f_1''} & & \downarrow{\scriptstyle f_1'} & & \downarrow{\scriptstyle f_1} \\ Y_1' \times_{Y_1} Y_1' & \overset{p_1}{\underset{q_1}{\rightrightarrows}} & Y_1' & \xrightarrow{g_1} & Y_1 \end{array}$$

and the same arguments to get a well defined map $c \cap - : \mathrm{CH}_k(Y_1) \to \mathrm{CH}_{k+p}(X_1)$ as before.

Next, we have to check conditions (1), (2), and (3) of Definition 0B76 for c. For example, suppose that $t : Y_2 \to Y_1$ is a proper morphism of schemes locally of finite type over Y. Denote as above the base changes of the first diagram to Y_1, resp. Y_2, by subscripts $_1$, resp. $_2$. Denote $t' : Y_2' \to Y_1'$, $s : X_2 \to X_1$, and $s' : X_2' \to X_1'$ the base changes of t to Y', X, and X'. We have to show that

$$s_*(c \cap \beta_2) = c \cap t_*\beta_2$$

for $\beta_2 \in \mathrm{CH}_k(Y_2)$. Choose $\beta_2' \in \mathrm{CH}_k(Y_2')$ with $g_{2,*}\beta_2' = \beta_2$. Since c' is a bivariant class and the diagrams

$$\begin{array}{ccc} X_2' & \xrightarrow{h_2} & X_2 \\ {\scriptstyle s'}\downarrow & & \downarrow{\scriptstyle s} \\ X_1' & \xrightarrow{h_1} & X_1 \end{array} \quad \text{and} \quad \begin{array}{ccc} X_2' & \xrightarrow{f_2'} & Y_2' \\ {\scriptstyle s'}\downarrow & & \downarrow{\scriptstyle t'} \\ X_2' & \xrightarrow{f_1'} & Y_1' \end{array}$$

are cartesian we have

$$s_*(c \cap \beta_2) = s_*(h_{2,*}(c' \cap \beta_2')) = h_{1,*}s'_*(c' \cap \beta_2') = h_{1,*}(c' \cap (t'_*\beta_2'))$$

and the final expression computes $c \cap t_*\beta_2$ by construction: $t'_*\beta_2' \in \mathrm{CH}_k(Y_1')$ is a class whose image by $g_{1,*}$ is $t_*\beta_2$. This proves condition (1). The other conditions are proved in the same manner and we omit the detailed arguments. □